

Executive Summary: Digital Concierge & Wayfinding Solution

This executive summary outlines the strategic implementation of the "Digital Concierge," an AI-enabled indoor navigation solution designed to streamline the transition to the new headquarters. By leveraging **Azure Maps** as the core mapping engine, this solution transforms static architectural diagrams and employee rosters into a dynamic, interactive wayfinding system.

1. Executive Overview

Moving to a new headquarters presents a significant change management challenge. As employees navigate new layouts, find colleagues, and locate essential services (like the Tech Hub or Mother's Room), "wayfinding friction" can lead to lost productivity and frustration.

The proposed solution is a **Digital Concierge**: a web-based application that combines high-fidelity indoor mapping with Natural Language Processing (NLP). Unlike a static directory, this tool understands *intent*. If an employee asks, "I need a new mouse," the system does not just search for the word "mouse"; it intelligently routes them to the **IT Genius Bar on Floor 1 (Room 1W-TechHub)**, provides walk-time estimates, and offers a booking link.

By anchoring this solution in **Azure Maps**, the organization capitalizes on enterprise-grade security, scalability, and the native "Azure Maps Creator" feature set designed specifically for smart spaces.

2. Solution Features & User Experience

The solution is divided into three functional layers: The **Map Interface**, the **Smart Search (NLP)**, and the **Directory Integration**.

A. The "Smart" Map (Azure Maps Creator)

The core visual experience is powered by Azure Maps.

- **Dynamic Routing:** The system calculates the shortest walkable path between two points (e.g., "My Desk" to "Cafeteria"), accounting for walls, elevators, and

stairwells.

- **Multi-Floor Navigation:** The map handles the complexity of vertical travel. It will explicitly instruct users: *"Walk to the central elevator bank, take the elevator to Floor 1, then turn left toward the East Wing."*
- **Points of Interest (POI) Overlay:** Critical zones identified in your floor plans—such as the **Training Centers (1W-Training)**, **Cafe (1E-Cafe)**, and **Executive Wing (6E-Exec)**—are permanently highlighted with interactive icons.

B. Natural Language "Intent" Search

We move beyond keyword matching. The system uses an NLP engine (via Azure OpenAI) to interpret user needs:

- **Service Discovery:**
 - *User:* "Where can I warm up my lunch?"
 - *System:* Identifies intent "Pantry/Kitchen." Locates the nearest pantry on the user's current floor or routes them to the **Main Cafeteria (1E-Cafe)**.
- **Departmental Routing:**
 - *User:* "I have a meeting with HR."
 - *System:* Recognizes "HR" is located in the **Executive/HR Wing (Floor 6, East)** and provides directions to the secure lobby.

C. The "Live" Roster

The system integrates directly with your existing seating data.

- **Dynamic Desk Finding:** When a user searches "Jason Maddix," the system queries the roster, retrieves the seat ID (1W-002), and drops a pin at that exact coordinate on the Floor 1 map.
- **Team Zones:** Users can search for teams (e.g., "Marketing Team") and see a heat map of where that department is clustered.

3. Technical Architecture & Stack

The architecture is designed to remain within the Microsoft Azure ecosystem, ensuring data privacy (GDPR/compliance) and seamless integration with existing IT infrastructure (Entra ID/Active Directory).

Component	Technology	Role in Solution
Map Engine	Azure Maps (Creator)	The heart of the system. It ingests CAD/DWG files to create the topology (walls, corridors) and renders the interactive map tiles.
Backend API	Python (FastAPI) on Azure App Service	Handles logic. It receives the user's query, consults the database, and asks Azure Maps for a route.
AI / NLP	Azure OpenAI Service (GPT-4o)	The "Brain." It translates natural language inputs ("I need IT help") into structured database queries (<code>SELECT * FROM Services WHERE Type='IT'</code>).
Database	Azure SQL Database	Stores the "meta-layer": the link between physical Room IDs (from the map) and logical data (Employee Names from your CSV, Service Descriptions).
Frontend	React / TypeScript	A lightweight, mobile-responsive web app. No app store download required; accessible via browser or QR codes posted in elevators.

4. Proof of Concept (POC) Scope

To validate value quickly, the POC will not map the entire building immediately. It will focus on a "Vertical Slice" that demonstrates the hardest technical challenges: **Service Routing** and **Desk Finding**.

Scope Boundary

- **Floors Involved:**
 - **Floor 1 (The Service Hub):** Selected because it contains the highest density of POIs (Cafe, Tech Hub, Training).
 - **Floor 2 (The Workspace):** Selected to demonstrate high-density cubicle navigation.
- **Data Set:**
 - The 647-person roster provided (filtered to occupants of Floors 1 & 2).
 - The "Hidden Services" list we generated (Tech Hub, Shipping, etc.).

POC Success Criteria

1. **Successful Vectorization:** Convert Floor 1 & 2 JPGs into a format Azure Maps can read (DWG/IMDF).

2. **"Blue Dot" Simulation:** Ability to click "Start Here" (e.g., Lobby) and see a polyline path drawn to a destination.
3. **Roster Resolution:** Typing "Jennifer Thomson" must successfully drop a pin at desk 1W-003.
4. **Intent Resolution:** Typing "Help, my laptop is broken" must route the user to 1W-TechHub.

5. Implementation Roadmap

Phase 1: Data Preparation (Weeks 1-2)

- **Digitization:** This is the most critical step. Azure Maps cannot route users through a JPG image. We must trace the walls and paths of the Floor 1 and Floor 2 images into **DWG (CAD) format**.
Deliverable: A "Manifest" file for Azure Maps defining walkability zones.
- **Data Cleaning:** Import the CSV roster into Azure SQL. Standardize the `Space Number` column to match the Room IDs in the CAD file.

Phase 2: The Azure Maps Setup (Weeks 3-4)

- Provision the **Azure Maps Creator** resource.
- Upload the converted CAD files to create the "Dataset" and "Tileset."
- Define the "Ontology" (tagging specific polygons as "Office," "Restroom," "Elevator").

Phase 3: The "Concierge" Logic (Weeks 5-6)

- Develop the Python backend.
- Connect the **Azure OpenAI** instance. Prompt engineer the system with the "Facilities Context" (teaching it that *Shipping = Dock 1E-021*).
- Build the simple React frontend.

Phase 4: User Testing (Week 7)

- Deploy to a test URL.
- Test specific "edge case" routes (e.g., routing from Floor 2 Desk to Floor 1 Cafe).

6. Strategic Recommendations

1. **QR Code Strategy:** Post "You Are Here" QR codes at key decision points (Elevator lobbies, Cafeteria entrance). The URL in the QR code should carry a payload (e.g., `?start_location=elevator_bank_floor_2`) so the map automatically knows the user's starting point.
2. **Kiosk Mode:** The React application can be deployed on large touchscreens in the main lobby as a permanent "Digital Directory."
3. **Maintenance:** The beauty of this architecture is that when people move, you only update the CSV/Database. You do **not** need to redraw the map. The map simply queries the database for the new location of "Jason Maddix."

Summary

This solution leverages your existing assets—high-quality floor plans and accurate seating data—and elevates them using Azure's spatial intelligence. By moving from a static "Directory" to a dynamic "Concierge," we ensure that the modern workplace experience begins the moment an employee steps through the door.